



PROFESSIONAL RESUME

MUHAMMAD IRFAN BIN AZMAN

NAME : MUHAMMAD IRFAN BIN AZMAN
POSITION : PIPING ENGINEER / JR PIPE STRESS ENGINEER
DATE OF BIRTH :
NATIONALITY : MALAYSIAN
MARITAL STATUS :
CONTACT ADDRESS :
Tel :
EMAIL :

EXPERIENCE SUMMARY

Piping Engineer in the oil & gas sector with 4 years of experience (including 2 years in piping engineering), bringing practical expertise in piping design, 3D modelling, Pipe Stress and layout creation. Skilled in interpreting P&IDs; and isometric drawings, with knowledge of ASME B31.3 and related industry codes. Proficient in Smart 3D, Navisworks, and AutoCAD for developing and validating piping systems that meet project specifications. Experienced in leveraging Power BI daily for data analysis, dashboard creation, and performance monitoring to support project decision-making. Holds a BEng in Mechanical Engineering from UTHM, with a strong focus on safety, quality, and continuous professional growth in dynamic engineering environments.

ACADEMIC/TECHNICAL QUALIFICATIONS:

2018 – 2020

Bachelor's Degree in Mechanical Engineering

Universiti Tun Hussein Onn, Malaysia

2014 – 2017

Diploma in Mechanical Engineering

Universiti Tun Hussein Onn, Malaysia

WORKING EXPERIENCE

Year : 2024 – Present

Company : Ranhill Worley

Position : Piping Design / Jr Pipe Stress Engineer

Project Involved:

- SEATRIUM / PETROBRAS, DED P82 FPSO, Kuala Lumpur

Job Description:

- Jr Pipe Stress Engineer
- Perform CEASER II stress modelling in accordance with ASME B31.3
- Extracted loads (Hanger, Nozzle, Restrain) from CESSER II for pipe support design
- Coordinated with stress engineer to implement piping flexibility solutions and modify routing/supports during review cycle
- Identified overstress and support clashes, proposing revisions for compliance with project standard

- Cross-checked pipe stress reports against 3D models and isometric drawings, marking up supports to align with the latest stress report
- ISOMETRIC Coordinator
- Coordinated and managed the issuance of isometric drawings for piping systems, ensuring compliance with project specifications and mechanical design standards.
- Reviewed and verified isometric drawings for accuracy against engineering requirements, maintaining data integrity throughout the project lifecycle
- Monitored and enforced strict weekly deadlines for drawing issuance, supporting project schedule adherence.
- Maintained detailed documentation of drawing revisions and re-issuance timelines for internal tracking and audits.
- Supported Lead and Senior Engineers with data consistency checks across design deliverables.
- Developed and presented interactive dashboards in Power BI to provide clients with real-time project insights, drawing progress, and data summaries.
- Reviewed P&ID; and 3D models to identify line changes against issued and pending isometrics.
- Assisted in preparing tubing MTO (Material Take-Off) to support procurement and construction activities.

Year : 2023 – 2024

Company : Ranhill Worley

Position : Graduate Proposal Engineer

Project Involved:

- Upstream oil & gas packages (multiple proposals)

Job Description:

- Prepared complete technical and commercial proposals for upstream oil & gas packages valued up to RM 15 million.
- Interpreted client RFQs and applied relevant codes to produce compliant equipment lists and technical data sheets.
- Developed detailed cost estimates and cash flow curves to support bid evaluations
- Coordinated multi-discipline inputs (engineering, procurement, tax, legal, treasury), ensuring 95% of submissions delivered on or ahead of schedule.
- Led both technical (execution plans, project team structures) and non-technical activities to ensure comprehensive proposal coverage.
- Consolidated inputs from diverse stakeholders into clear, competitive submissions.
- Contributed to securing two project wins (CPOC FEED & Shell MDS reimbursable contract basis).

Year : 2022 – 2023

Company : Trengco Energy (M) Sdn Bhd

Position : Project Engineer

Project Involved:

- PETRONAS Penapisan Terengganu (PPTSB): Cleaning & desludging, waste treatment, and civil works.
- PETRONAS Chemicals Olefins, Glycols & Derivatives (PCOGD): Amine pump replacement.
- Arkema Thiochemicals Kerteh: Fabrication and installation of actuated valve supports.

Job Description:

- Planned, executed, and controlled projects in compliance with quality and safety standards
- Analyzed engineering drawings, specifications, and purchase orders to ensure project requirements were met
- Prepared execution documentation including work instructions and operating procedures.
- Supported the Project Manager in negotiating change orders, coordinating subcontractors, and ensuring smooth project execution.
- Conducted site visits during pre-bid stage to evaluate project scope and risks.
- Organized cross-functional meetings to resolve issues, align stakeholders, and maintain project control.

Year : 2021 – 2022

Company : PETRONAS Gas Berhad

Position : Mechanical Rotating GEES

Project Involved:

- Utilities Kerteh – GE MS 6001B GT
- Utilities Kerteh – MRO Pump

Job Description:

- Prepare pre-emptive work control sheet for boiler feed water pump, cooling tower (gearbox & fan blade)
- Involve in Major Inspection (MI) for gas turbine (GT-B) at utilities Kerteh – GE MS 6001B GT

- Involve in refurbishment work for MRO Pump at Utilities Kerteh

Year : 2020 – 2021

Company : TMA Construction

Position : Asst Project Engineer

Project Involved:

- EPCC NAP (New Air Compressor Project) at PGB (GPP 5&6)

Job Description:

- Plan and coordinate in site activity for EPCC NAP (New Air Compressor Project) at PGB (GPP 5&6), earthwork and site preparation.
- Prepare daily progress report for main contractor (CMI)
- Prepare and review document for site inspection.
- Attend site visit prior project bidding.

OTHERS (TRAININGS/SKILLS/etc.)

- Intergraph Smart 3D
- CAESER II
- AutoCAD
- Power BI
- SolidWorks
- Navisworks
- Microsoft Office
- Pipedata-Pro

INVOLVEMENTS

2020 Board of Engineer (BEM), Graduate Engineer Mechanical – G1199436A

2020 IEM, Member of Institutes of Engineer Malaysia